Revision and consolidation

Lecture and worked examples

Revision strategy

Effective revision alternates retrieval from memory, explicit correction of mistakes and reuse in a problem. Passively rereading the course is not enough: you must attempt a question without a model, compare your answer, then redo the question after a delay.

Diagnosis of what has been learned

- Identify what has been learned.
- Classify the mistakes.
- Plan the remediation.

Diagnosis of what has been learned

Answer the following 14 questions without notes, two per chapter. Record each error in a grid: knowledge, choice of method, calculation, reading, written presentation. Then choose two personal priorities.

1. Sequences. The sequence (u_n) is arithmetic with first term $u_0 = 7$ and common difference -3. Calculate u_{10}. …
2. Sequences. The sequence (v_n) is geometric with first term $v_0 = 200$ and common ratio 1.05. Calculate v_2. …
3. Exponential functions. State the direction of variation of $f(x) = 3 \times 0.8^x$ on $[0, +\infty)$. …
4. Exponential functions. A quantity of 500 increases by 4% per year. Write the model $f(x)$ after x years. …
5. Common logarithm. Solve $10^x = 0.01$. …
6. Common logarithm. Calculate $\log(1000) - \log(10)$. …
7. Reciprocal function. Solve $\dfrac{1}{x} = 4$ for $x \neq 0$. …

Worked answer: Diagnosis of what has been learned

1. -23 · **2.** 220.5 · **3.** decreasing · **4.** 500×1.04^{x} · **5.** $x = -2$ · **6.** 2 · **7.** 0.25

Diagnosis of what has been learned

8. Reciprocal function. State the direction of variation of $x \mapsto \dfrac{1}{x}$ on $(0, +\infty)$. ...
9. Two variables. With the fit $y = 2x + 3$, estimate y for $x = 10$. ...
10. Two variables. Give the mean point of the series x: 1, 2, 3 and y: 4, 6, 8. ...
11. Conditional probability. $P(A) = 0.4$ and $P_A(B) = 0.5$. Calculate $P(A \cap B)$. ...
12. Conditional probability. $P(A \cap B) = 0.12$ and $P(A) = 0.3$. Calculate $P_A(B)$. ...
13. Binomial distribution. X follows the distribution $\mathcal{B}(3, 0.5)$. Calculate $P(X = 3)$. ...
14. Random variable. X takes the values 0, 1, 2 with probabilities 0.5, 0.3, 0.2. Calculate $E(X)$. ...

Worked answer: Diagnosis of what has been learned

8. decreasing · **9.** 23 · **10.** $G(2, 6)$ · **11.** 0.2 · **12.** 0.4 · **13.** 0.125 · **14.** 0.7

Diagnosis of what has been learned

The diagnosis is used to choose the remediation: recognise the mistake, recall the useful rule, redo a similar question, then check without help.

Diagnosis of what has been learned

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Diagnosis of what has been learned

1 h session: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.

2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.

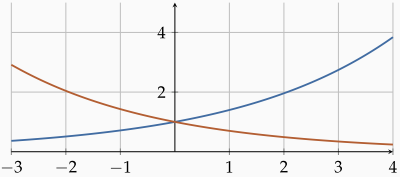
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Reactivation: exponential functions

- Reactivate exponential models.
- Solve and interpret.
- Check the units.

Reactivation: exponential functions

A value of 18,000 increases by 3.5% per year. Write the model, calculate the value after eight years and determine when it will exceed 25,000.



Reactivation: exponential functions

An exponential function models a constant relative change. First identify the initial value and the multiplier.

Reactivation: exponential functions

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Reactivation: common logarithm (base 10)

- Reactivate logarithms.
- Solve an equation.
- Check the conditions.

Reactivation: common logarithm (base 10)

Solve $10^x = 340$, $1.06^n = 2$ and $1.06^n \geq 2$. For each answer, distinguish the real value, the approximate value and the minimum integer index.

Reactivation: common logarithm (base 10)

The logarithm turns an exponential equation into an equation in the exponent. Conditions: base positive and different from 1, right-hand side positive.

Reactivation: common logarithm (base 10)

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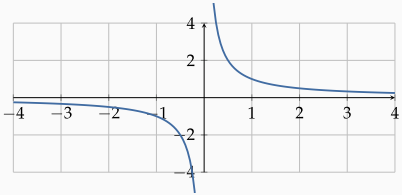
Reactivation: reciprocal function

- Reactivate the reciprocal function.
- Carry out a study.
- Sketch and interpret.

Reactivation: reciprocal function

Study $f(x) = x + \dfrac{16}{x}$ on $]0, +\infty[$ and interpret its minimum in an average-cost context. Produce the table of variations and a sketch.



Reactivation: reciprocal function

For a function containing $\frac{1}{x}$, start by excluding $x = 0$, then differentiate, factorise and study the sign before the variations.

Reactivation: reciprocal function

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